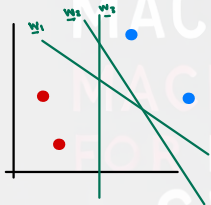


Week 7, Support vector machines

CAPP 30254, Machine Learning for Public Policy

Extreme example:



all perfect classifiers!
but they do not have the same hinge loss
is there a "best"?

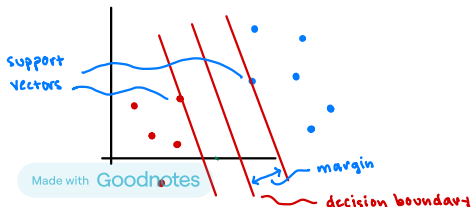
Support vector machines

The model whose loss function is the hinge loss with a ridge term:

$$\ell(\mathbf{w}) = \underbrace{C}_{\text{ridge penalty}} \sum_{i=1}^n \underbrace{\max(0, 1 - y_i \cdot \hat{y}_i)}_{\text{hinge loss}} + \underbrace{\frac{1}{2} \|\mathbf{w}\|^2}_{\text{ridge regularization}}$$

is called a *support vector machine*.

When trained with a linearly separable training set, the weight vector $\mathbf{w}_{SVM}$ is the weight vector with the maximum possible margin between classes.



Example

Consider the following training data in $\mathbb{R}^2$:

$$-1 : (1, 4), (2, 3)$$

$$+1 : (6, -1), (7, -1), (5, -3)$$

Can SVM be trained with this dataset? What are the support vectors? What is the decision boundary?

use graph paper and inspection

Slack variables

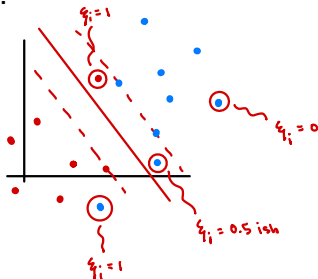
Real-world data has:

- noise
- outliers
- overlapping data

When the data isn't perfectly separable, we can introduce *slack variables* ξ_i which allow some points to be within the margin or misclassified in exchange for a better overall fit.

Slack variables ξ_i are nonzero for training examples i that are within the margin or misclassified.

- sample i
- $\xi_i = 0$: correctly classified, outside the margin
 - $0 < \xi_i < 1$: correctly classified, inside the margin
 - $\xi_i = 1$: misclassified



Soft-margin SVM

previous: "Hard-margin" SVM

An SVM that allows slack variables is called a *soft-margin* SVM. The soft-margin SVM loss function is given by:

$$\ell(\mathbf{w}) = C \sum_{i=1}^n \xi_i + \frac{1}{2} \|\mathbf{w}\|^2$$

hyperparameter penalize violations

C: controls tradeoff between margin width and misclassification tolerance

Large C:

- high penalty on misclass.
- closer to hard-margin

Small C:

- more tolerance for misclass.
- wider margin

compromise between a simple model (wide margin) and perfect fit for training data

Example

Say we add two more samples to our previous training data:

$-1 : (1, 4), (2, 3), (4, 3)$ $\xi_i = 0.25$

$+1 : (6, -1), (7, -1), (5, -3), (3, 1)$ $\xi_i = 0$

rest: $\xi_i = 0$

and keep the same decision boundary and support vectors as before. What are the values ξ_i for each training sample?

^
approximate

By inspection only

What if we made C very big?

move the boundary,
if possible



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¹Generated by GPT-4, DALL-E 3 after the prompt: “Hi, could you please make a minimalist logo for a machine learning class...”